CS 132 Capstone Presentation Defense Master Guide

Complete Methodological & Statistical Rationale: "What It Does & Why It Was Used"

Team FYI – FOI Your Information · UP Diliman Department of Computer Science
Project: **Who Gets an Answer? eFOI Response Times and Outcomes in 2025**
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Theme: **Pilipinas in a Nutshell (PiNut) 2026** · UN SDG 16 (Peace, Justice & Strong Institutions)

How to Use This Master Manual

This manual is specifically engineered for your capstone presentation defense. In data science defenses, professors don't just ask **what your findings are**; they probe **why you chose a specific statistical test over standard alternatives**, **how that test works internally**, and **whether your data satisfies its theoretical assumptions**.

For every single method, test, and analytical decision in your project, this document details:

1. ⚙️ **What It Does:** The inner mathematical mechanism, inputs, and outputs explained in plain English.
2. 🎯 **Why This Specific Test Was Used:** Direct justification against textbook alternatives (e.g., ANOVA, t-tests, K-S tests, logistic regression, simple random sampling).
3. 🗣️ **Quick Spoken Defense:** An authoritative 15-to-30-second script you can say immediately.
4. 🔬 **Exact Numbers & Formulas:** Test statistics, degrees of freedom, p-values, effect sizes, and formulas.
5. ⚠️ **Professor Traps:** Anticipated "gotcha" questions from skeptical panelists and how to defuse them.

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1. Executive Narrative & Core Story

The Central Thesis: "A Fast Reply is Not a Yes"

🗣️ QUICK SPOKEN DEFENSE (What to Say Out Loud)

"Our central finding is that **'A fast reply is not a yes.'** While agency groups differ drastically in initial speed and disclosure rates, speed itself does not predict success. In fact, denied requests received a faster initial reply (weighted median 1.0 day) than successful ones (1.5 days). Furthermore, initial response speed is often just an automated or administrative acknowledgment, masking systemic issues like the complete silence observed in over a quarter of Water District requests."

⚙️ WHAT IT DOES (Mechanism & Logic)

Synthesizes the findings of three separate empirical investigations:
1. **RQ1:** Measured the speed of first contact across 6 official agency groups ($H = 218.71, p = 2.80 \times 10^{-45}$).
2. **RQ2:** Audited the final disclosure outcomes across the full 2025 population ($\chi^2 = 134.77, p = 2.32 \times 10^{-27}$).
3. **RQ3:** Evaluated whether response speed correlates with disclosure success ($p = 0.989$; dominance = 0.500).

🎯 WHY THIS SPECIFIC TEST/METHOD WAS USED (vs. Alternatives)

- **Context:** Government compliance reports under Executive Order No. 2, s. 2016 celebrate aggregate speed compliance (meeting the 15-day ceiling).
- **Why this narrative was chosen:** Our microdata audit reveals that speed is decoupled from transparency. Focusing exclusively on response time creates a perverse incentive for agencies to issue rapid, boilerplate denials or referrals rather than substantive disclosures.

2. Data Collection & Scraping Architecture

Method 1: Playwright with Google Chrome via Remote Debugging (CDP)

⚙️ WHAT IT DOES (Mechanism & Logic)

Launches a live Google Chrome process with a dedicated user profile (`.chrome-profile/`) listening on remote debugging port `9222`. Playwright connects to this existing browser instance using the Chrome DevTools Protocol (CDP) to drive navigation, execute JavaScript, and retrieve rendered DOM trees.

🎯 WHY THIS SPECIFIC TEST/METHOD WAS USED (vs. Alternatives)

- **Why NOT `requests`, `urllib`, or `BeautifulSoup`:** The eFOI portal sits behind Cloudflare bot mitigation. Plain HTTP clients receive an immediate **HTTP 403 Forbidden** challenge page regardless of custom User-Agent headers.
- **Why NOT standard headless Playwright/Selenium:** Headless automation browsers leak browser fingerprints (e.g., `navigator.webdriver = true`, missing WebGL vendor strings) and fail Cloudflare Turnstile challenges after page 1.
- **Why live Chrome via CDP:** A genuine Chrome instance with a persistent profile retains legitimate TLS ciphers, session cookies, and storage, passing Cloudflare without automated bypasses. If an interactive challenge appears, team members solve it manually.

Method 2: Binary Search for Pagination Boundaries

⚙️ WHAT IT DOES (Mechanism & Logic)

Executes a divide-and-conquer binary search algorithm over the integer listing page URLs (`/requests/page/N/`) to locate the exact page indices where calendar year 2025 filings start and end.

🎯 WHY THIS SPECIFIC TEST/METHOD WAS USED (vs. Alternatives)

- **Why NOT linear scraping from Page 1:** The portal contains over 40,000 historical requests across thousands of pages. Linear scraping from page 1 would require thousands of unnecessary HTTP requests, imposing abusive server load.

- **Why Binary Search works:** The eFOI portal enforces strict chronological ordering (newest filings on page 1, oldest on the last page). This monotonic ordering makes binary search run in $O(\log M)$ time, pinpointing the 2025 boundary (pages 1024 to 2002; 979 pages total) in under 15 HTTP requests.

Method 3: Ground-Truth Manual Audit (100 Random Samples)

⚙️ WHAT IT DOES (Mechanism & Logic)

Extracts 100 randomly sampled rows (`validation_100.csv`) from the final processed dataset. Human evaluators independently open each live portal URL and verify whether tracking number, filing timestamp, first reply timestamp, and outcome status match the parsed CSV fields.

🎯 WHY THIS SPECIFIC TEST/METHOD WAS USED (vs. Alternatives)

- **Why NOT trust unit tests alone:** Automated tests on mock HTML fixtures only verify parser logic against assumed HTML structure. Live web portals frequently change DOM structures, introduce multiline white-space, or format dates inconsistently.
- **Result:** We achieved a **100% concordance rate (100/100)** across all audited fields, providing empirical proof of zero scraper-induced measurement error.

3. Sampling Design & Survey Weighting

Method 4: Disproportionate Stratified Random Sampling

⚙️ WHAT IT DOES (Mechanism & Logic)

Divides the population into 24 distinct strata defined by **Agency Group (6)** × **Filing Quarter (4)** (seed 132). It caps large groups at 300 requests (distributed proportionally across quarters) and draws a full census for groups with fewer than 300 requests in 2025.

🎯 WHY THIS SPECIFIC TEST/METHOD WAS USED (vs. Alternatives)

- **Why NOT Simple Random Sampling (SRS):** The 2025 population ($N = 14,668$) is severely imbalanced: National Government Agencies (NGAs) represent **83.3%** ($N = 12,219$), while Local Governments (LGUs) represent only **1.2%** ($N = 175$) and Water Districts **2.0%** ($N = 287$). In an SRS of 1,600 requests, LGUs would receive only ≈ 19 samples —making subgroup medians and non-parametric tests statistically powerless.
- **Why NOT Proportional Stratified Sampling:** Proportional allocation preserves the same skewed ratios as SRS, failing to protect small subgroups.
- **Why Disproportionate Stratification:** Allocating ≈ 300 samples per group guarantees small standard errors (≤ 0.03) and sufficient degrees of freedom across all institutional types, while taking a full census of LGUs ($N = 175$) and Water Districts ($N = 287$) eliminates sampling error entirely for those groups.

Method 5: Post-Stratification Survey Weighting ($w_h = \frac{N_h}{n_{\text{collected}}}$)

⚙️ WHAT IT DOES (Mechanism & Logic)

Assigns each sampled request an analytic weight equal to the inverse of its stratum selection probability. In any pooled calculation (mean, median, quartile, or test), each observation is multiplied by its weight so that small oversampled groups don't dominate national estimates.

🎯 WHY THIS SPECIFIC TEST/METHOD WAS USED (vs. Alternatives)

- **Why survey weights are mandatory:** Without weights, LGUs (10.7% of the sample) would carry 9 times more influence than their true population share (1.2%), while NGAs (18.2% of the sample) would be underweighted by a factor of 4.5.
- **Why adjust for collected count instead of drawn count:** Of 1,663 drawn requests, 22 detail pages had vanished from the portal (20 in Law Enforcement Q1, 2 in NGA Q1).
- **Why NOT replace vanished pages with new draws:** 77.3% of the vanished pages were listed as unsuccessful. Drawing replacement pages from live records would introduce **survivorship bias**, artificially inflating transparency.
- **The Formula:** By defining $w_h = \frac{N_h}{n_{\text{collected}}}$ rather than $\frac{N_h}{n_{\text{drawn}}}$, the weights of the remaining collected requests in those strata expand to accurately sum to the exact population of $14,668$ without introducing replacement bias.

Method 6: Dedicated Missing Value Encoding for Silence (No Imputation)

⚙️ WHAT IT DOES (Mechanism & Logic)

Preserves `first_response_days` as blank (`NaN`) for the 205 sampled requests where the agency never sent a message. Evaluates response times conditionally on replied requests, while treating the no-reply share as an independent categorical variable.

🎯 WHY THIS SPECIFIC TEST/METHOD WAS USED (vs. Alternatives)

- **Why NOT Mean or Median Imputation:** Imputing a central tendency fabricates a fictitious government action that never took place, distorting variance and misleading researchers.
- **Why NOT Zero Imputation:** Imputing 0 implies the agency responded in 0.00 seconds (instantaneous compliance), the exact opposite of reality (the agency ignored the citizen).
- **Why NOT Listwise Deletion:** Dropping silent requests from the entire study would hide the fact that over a quarter of Water District requests vanish into institutional silence.

4. Normality Testing & Distributional Diagnostics

Method 7: The Shapiro–Wilk Test of Normality

⚙️ WHAT IT DOES (Mechanism & Logic)

Tests the null hypothesis (H_0) that a sample comes from a normal distribution. It sorts the sample values ($x_{(1)} \leq x_{(2)} \leq \ldots \leq x_{(n)}$) and calculates the ratio of the best linear unbiased estimator of the slope to the sample variance, producing the test statistic $W \in [0, 1]$:

$$W = \frac{\left(\sum_{i=1}^{n} a_i x_{(i)}\right)^2}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

A perfect normal bell curve yields $W = 1.0$. As the sample deviates from a bell curve, W drops toward 0. If $p < 0.05$, reject normality.

🎯 WHY THIS SPECIFIC TEST/METHOD WAS USED (vs. Alternatives)

- **Why NOT Kolmogorov–Smirnov (K-S):** The classical K-S test is designed for continuous distributions with **known** population mean and variance. When μ and σ are estimated from the sample, K-S is invalid unless using Lilliefors correction, and even then has notoriously weak power.
- **Why NOT Anderson–Darling or D'Agostino-Pearson:** D'Agostino only tests skewness and kurtosis. Shapiro–Wilk evaluates the entire regression of order statistics against normal scores, giving it the highest empirical power for sample sizes under a few thousand.
- **Why test within each agency group separately:** ANOVA requires that residuals **within each group** are normal. Testing the pooled sample would be invalid because different group means create artificial multimodality.
- **Why test log-transformed data:** To verify whether a standard monotonic transformation could rescue parametric assumptions. Normality was decisively rejected even after log-transforming ($p < 0.002$ in all 6 groups).

⚠️ PROFESSOR TRAP & SKEPTIC CHALLENGE

The Professor Trap: "Shapiro–Wilk is overpowered for $n \approx 300$. It will reject normality for trivial deviations!" **Your Answer:** "We did not rely on p-values alone. We looked at the W test statistic itself. A normal distribution has $W \approx 1.0$. Our raw W values were **between 0.21 and 0.57** (Law Enforcement was $W = 0.215$). That is an extreme, structural departure from normality. Skewness was **+4.82**, and the mean (17.4d) was 10 times the median (1.58d). Non-normality is a physical reality in this data, not a sample-size artifact."

Agency Group	n	Raw W	Raw p-value	Log W	Log p-value
National Agencies (NGA)	281	0.312	4.4×10^{-31}	0.982	0.0013
Gov't Corporations (GOCC)	260	0.414	3.0×10^{-28}	0.969	2.0×10^{-5}
State Universities (SUC)	260	0.497	1.5×10^{-26}	0.963	3.3×10^{-6}
Water Districts	212	0.394	3.8×10^{-26}	0.965	4.3×10^{-5}
Local Governments (LGU)	145	0.570	7.6×10^{-19}	0.968	0.0019
Law Enforcement (LEA)	278	**0.215**	1.6×10^{-32}	0.954	1.1×10^{-7}

Method 8: Quantile–Quantile (Q-Q) Probability Plots

⚙️ WHAT IT DOES (Mechanism & Logic)

Plots the sorted empirical quantiles of the sample against the theoretical quantiles of a standard normal distribution. If the data is normally distributed, all points fall along a straight diagonal $45°$ reference line.

🎯 WHY THIS SPECIFIC TEST/METHOD WAS USED (vs. Alternatives)

- **Why NOT Histograms alone:** Histograms depend heavily on arbitrary bin-width selections, which can obscure heavy-tail behavior or over-smooth discrete clusters.
- **Diagnostic Value:** Q-Q plots graphically expose tail weight. In our data, the points curve sharply upward on the right tail, visually proving that extreme slow replies (up to 527 days) represent severe heavy-tailed non-normality.

5. Research Question 1: Speed Across Agency Groups

Method 9: Kruskal–Wallis One-Way Non-Parametric ANOVA

⚙️ WHAT IT DOES (Mechanism & Logic)

Tests the null hypothesis (H_0) that k independent groups originate from the same continuous distribution of response times. It pools all 1,436 replied requests, replaces raw elapsed days with global ranks (1 to $1,436$), and computes the test statistic H:

$$H = \frac{12}{N(N+1)} \sum_{j=1}^{k} \frac{R_j^2}{n_j} - 3(N+1)$$

where R_j is the sum of ranks in group j. Under H_0, H asymptotically follows a chi-square distribution with $k-1=5$ degrees of freedom.

🎯 WHY THIS SPECIFIC TEST/METHOD WAS USED (vs. Alternatives)

- **Why NOT One-Way ANOVA:** ANOVA assumes normally distributed residuals and homoscedasticity. With skewness of +4.82 and Shapiro–Wilk rejecting normality at $p < 10^{-18}$, ANOVA's Type I error rate becomes uncontrollable.
- **Why NOT Welch's ANOVA:** While Welch handles unequal variances, it still assumes normality and remains fragile to extreme right-tail outliers.
- **Why Kruskal–Wallis:** By replacing values with ranks, Kruskal–Wallis becomes completely immune to outlier distortion (e.g., a 527-day response becomes simply rank 1,436). It is the mathematically mandated non-parametric test.

🔬 EXACT NUMBERS, FORMULA & CITATIONS

- Test Statistic: $H = 218.71, \text{df} = 5, p = 2.80 \times 10^{-45}$.
- Effect Size: $\varepsilon^2 = \frac{H}{\frac{N^2-1}{N+1}} = \frac{218.71}{1435} = 0.152$ (moderate effect size).
- Decision: Decisively reject H_0 at $\alpha = 0.05$. Agency groups differ massively in initial speed ranks.

Method 10: Pairwise Mann–Whitney U Tests with Bonferroni Correction

⚙️ WHAT IT DOES (Mechanism & Logic)

Conducts post-hoc pairwise comparisons between all possible pairs of groups ($\binom{6}{2} = 15$ pairs). Ranks the observations in each two-group subset and computes the Mann–Whitney U statistic. To maintain the Family-Wise Error Rate (FWER) at $\alpha = 0.05$, each calculated p-value is multiplied by 15 ($p_{\text{bonferroni}} = p \times 15$).

🎯 WHY THIS SPECIFIC TEST/METHOD WAS USED (vs. Alternatives)

- **Why NOT Tukey's HSD:** Tukey's test assumes normal residuals and equal variance, making it invalid following a Kruskal–Wallis test.
- **Why Bonferroni:** With 15 pairwise comparisons, the unadjusted chance of at least one false positive is $1-(1-0.05)^{15} \approx 53.7\%$. Bonferroni provides the strictest, most conservative protection against Type I error.
- **Result: 11 of the 15 pairs remained statistically significant** after Bonferroni correction.

🔬 EXACT NUMBERS, FORMULA & CITATIONS

Weighted Median Wait Times:

1. **Law Enforcement (LEA):** 0.56 days (13.4 hours)
2. **Water Districts:** 1.12 days
3. **National Agencies (NGA):** 1.16 days
4. **Gov't Corporations (GOCC):** 2.65 days
5. **State Universities (SUC):** 3.24 days
6. **Local Governments (LGU):** 4.68 days

The 4 Non-Significant Pairs after Bonferroni: GOCC vs. Water District ($p = 1.0$), SUC vs. LGU ($p = 0.464$), GOCC vs. SUC ($p = 0.260$), NGA vs. Water District ($p = 0.202$).

Method 11: Chi-Square Test of Proportions for Institutional Silence

⚙️ WHAT IT DOES (Mechanism & Logic)

Evaluates whether the probability of receiving **zero agency response** differs across agency groups. Constructs a 6×2 contingency table (Agency Group $\times$ Replied vs. No Reply) and computes Pearson's chi-square statistic:

$$\chi^2 = 100.94, \text{df} = 5, p = 3.34 \times 10^{-20}$$

🎯 WHY THIS SPECIFIC TEST/METHOD WAS USED (vs. Alternatives)

- **Why this test was necessary:** Evaluating speed only on replied requests creates a severe survivorship bias. A group could appear "fast" simply because it answers 3 easy requests in 1 hour while ignoring the other 97.
- **Key Finding:** Water Districts had a fast replied median (1.12 days), but **26.1% of their sampled requests received no response whatsoever**, compared to only 0.7% in Law Enforcement. Silence is non-random and institutionally clustered.

6. Research Question 2: Outcome Proportions Across Groups

Method 12: Pearson's Chi-Square Test of Independence

⚙️ WHAT IT DOES (Mechanism & Logic)

Tests whether request outcome is independent of agency group across the full frame population ($N = 14,668$). Calculates the expected cell counts under the null hypothesis of independence:

$$E_{ij} = \frac{\text{Row Total}_i \times \text{Column Total}_j}{\text{Grand Total}}$$

and measures the squared standardized deviations:

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

🎯 WHY THIS SPECIFIC TEST/METHOD WAS USED (vs. Alternatives)

- **Why NOT Fisher's Exact Test:** Fisher's exact test is computationally intensive and designed for small cell counts ($n < 5$). In our 6×2 contingency table, the smallest expected cell count was **65.6** (far above the threshold of 5), satisfying the asymptotic requirements of Pearson's chi-square test.
- **Test Statistic:** $\chi^2 = 134.77, \text{df} = 5, p = 2.32 \times 10^{-27}$. Decisively reject H_0. Success rates differ significantly across government sectors.

Method 13: Cramér's V Effect Size

⚙️ WHAT IT DOES (Mechanism & Logic)

Converts the chi-square statistic into a normalized correlation metric between 0.0 (no association) and 1.0 (perfect association) that is completely independent of sample size:

$$V = \sqrt{\frac{\chi^2}{N \times \min(r-1, c-1)}} = \sqrt{\frac{134.77}{14668 \times 1}} = 0.096$$

🎯 WHY THIS SPECIFIC TEST/METHOD WAS USED (vs. Alternatives)

- **Why p-value alone is insufficient:** With $N = 14,668$, even a minuscule, meaningless difference of 1 percentage point will produce $p < 10^{-15}$.
- **Scientific Insight:** A Cramér's V of **0.096 indicates a weak association**. While agency groups differ systematically, institutional sector alone accounts for very little of the variance in disclosure. Specific request contents, legal exemptions, and record existence matter far more.

Method 14: Denominator Sensitivity Analysis (All Requests vs. Closed-Only)

⚙️ WHAT IT DOES (Mechanism & Logic)

Re-runs the chi-square test on two different operational definitions of the denominator:

1. **All-Requests ($N = 14,668$):** Denominator includes `Open` and `Referred` requests.
2. **Closed-Only ($n = 12,969$):** Denominator restricted strictly to terminal adjudications (`Successful`, `Partially Successful`, `Denied`, `Closed`).

🎯 WHY THIS SPECIFIC TEST/METHOD WAS USED (vs. Alternatives)

- **Why this method was used:** Administrative backlogs vary across agencies. An agency with many pending requests might look "unsuccessful" simply because it has not finished processing them.
- **Result:** Under closed-only requests, LGU success jumped from 74.9% to 91.0%, but the statistical verdict was identical ($\chi^2 = 123.88, p = 4.73 \times 10^{-25}, V = 0.098$). Group rankings remained stable, proving that backlog definitions do not alter our conclusions.

7. Research Question 3: Response Speed vs. Request Outcome

Method 15: Weighted Mann–Whitney Permutation Test (Monte Carlo Null)

⚙️ WHAT IT DOES (Mechanism & Logic)

Evaluates whether the distribution of `first_response_days` differs between successful and unsuccessful outcomes among closed requests ($n = 1,387$).

1. **Weighted Dominance Statistic:** Calculates the survey-weighted probability that a successful request waited longer than an unsuccessful one:

$$\text{Stat} = P(X_{\text{success}} > Y_{\text{unsuccess}}) + 0.5 \times P(X = Y)$$

   Under the null hypothesis of no relationship, this statistic equals exactly 0.500 (pure 50/50 chance).
2. **Monte Carlo Permutation:** Randomly shuffles the outcome labels across requests **9,999 times** (seed 132) while holding survey weights and response times fixed, constructing an exact empirical null distribution.
3. **Empirical p-value:** Counts how often permuted worlds produce a departure from 0.500 as large as observed.

🎯 WHY THIS SPECIFIC TEST/METHOD WAS USED (vs. Alternatives)

- **Why NOT standard `scipy.stats.mannwhitneyu`:** Standard rank-sum functions assume simple random sampling with equal weights. In our stratified design, an unweighted test would treat an LGU case ($w \approx 1.0$) as having equal population importance to an NGA case ($w \approx 41.0$), creating massive bias.
- **Why NOT Logistic Regression:** Logistic regression assumes a linear relationship between log-odds and response time, which is violated by the heavy-tailed, non-linear distribution of wait times.
- **Result:** Observed dominance was **0.500**, yielding $p = 0.989$. We completely fail to reject H_0. There is zero evidence that speed correlates with success in the population.

🔬 EXACT NUMBERS, FORMULA & CITATIONS

Weighted Median Wait Times:

- **Successful Requests:** 1.51 days ($n = 968$)
- **Unsuccessful Requests:** 1.01 days ($n = 419$)
- **Success rates across response-time quartiles:**
  - 0.0 to 0.6 days: 71.9%
  - 0.6 to 1.3 days: 67.7%
  - 1.3 to 3.9 days: 78.3%
  - 3.9 to 527 days: 67.1%

  (Spread is only 11.1 percentage points with zero monotonic trend).

Method 16: Within-Group Subgroup Consistency Testing

⚙️ WHAT IT DOES (Mechanism & Logic)

Executes independent Mann–Whitney tests between speed and success within each of the 6 individual agency groups.

🎯 WHY THIS SPECIFIC TEST/METHOD WAS USED (vs. Alternatives)

- **Why this test was used:** To rule out **Simpson's Paradox**—the phenomenon where a relationship exists within every subgroup but disappears when data is pooled together.
- **Result:** In every single agency group, the test remained non-significant ($p > 0.05$):
  - NGA ($p = 0.949$) | GOCC ($p = 0.911$) | SUC ($p = 0.366$)
  - Water Districts ($p = 0.674$) | LGU ($p = 0.401$) | LEA ($p = 0.108$)

  Speed and outcome are unlinked across every sector of government.

Method 17: Extreme Rank Imputation for Non-Response Sensitivity

⚙️ WHAT IT DOES (Mechanism & Logic)

Takes the 22 closed requests that never received an agency reply (all 22 of which were unsuccessful; 20 in Water Districts, 2 in GOCC), assigns them the **slowest possible rank** (placing them at the back of the queue), and re-runs the Mann–Whitney test.

🎯 WHY THIS SPECIFIC TEST/METHOD WAS USED (vs. Alternatives)

- **Why this test was used:** Skeptics could argue that leaving out silent requests biased RQ3 because being ignored is the ultimate form of being slow.
- **National Pooled Result:** The weighted test remains completely non-significant ($p = 0.914$), because 22 requests represent a tiny weighted fraction of national filings.
- **Water District Subgroup Result:** Within Water Districts alone, the test **flips from non-significant ($p = 0.674$) to statistically significant ($p = 0.0164$)**!
- **Scientific Takeaway:** For the nation as a whole, speed does not predict success. But for Water Districts specifically, being ignored is directly tied to failure.

8. Data Visualization & Nutshell Plot Mechanics

Method 18: Logarithmic Beeswarm / Strip Plot with Categorical Silence Column

⚙️ WHAT IT DOES (Mechanism & Logic)

Visualizes all 1,641 requests simultaneously:

1. **Logarithmic Horizontal Scale:** Compresses values from 0.001 days to 527 days onto an interpretable visual axis.
2. **Dedicated Categorical Column:** Allocates a distinct vertical band on the right for requests with 0 replies ($\log(0)$ is undefined).
3. **Outcome Color Mapping:** Blue = Successful/Partially Successful, Pink = Unsuccessful, Gray = Open/Referred.
4. **Vertical Jitter:** Applies uniform vertical offset to eliminate overplotting.

🎯 WHY THIS SPECIFIC TEST/METHOD WAS USED (vs. Alternatives)

- **Why NOT Bar Charts:** Bar charts only show aggregate means or medians, completely hiding the heavy-tailed skew, the cluster of same-day replies, and the 205 silent requests.
- **Why NOT Linear Scatter Plots:** On a linear scale, 85% of requests bunch into an indistinguishable vertical stripe against the 0-axis, making visual analysis impossible.
- **Why this design tells the complete story:** It allows viewers to see in a single glance that pink (denial) dots cluster just as heavily at the 1-day mark as blue dots, visually proving that **a fast reply is not a yes**.

9. Understanding p-Values & Professor Traps

What a p-Value Actually Implies (The "Fluke Meter")

⚙ WHAT IT DOES (Mechanism & Logic)

A p-value measures **the probability of observing your sample result purely by random luck, assuming there is zero real effect in reality (H_0)**.

- **The Coin Flip Analogy:** Flip a coin 10 times and get 5 Heads $\rightarrow p \approx 1.0$ (Expected; no evidence of trickery). Flip 10 Heads in a row $\rightarrow p = 0.001$ (Too weird to be luck; reject fair coin hypothesis).
- **Small p ($p < 0.05$):** Reject H_0. The observed pattern is statistically significant.
- **Large p ($p \geq 0.05$):** Fail to reject H_0. Data matches random noise.

Translation of Every p-Value in Your Project:

Test in Your Project	Exact p-value	What It Proves in Plain English
Shapiro–Wilk Normality	$p < 10^{-18}$	Chance of normal distribution is zero. Proves ANOVA cannot be used.
RQ1 Speed by Group	$p = 2.80 \times 10^{-45}$	Chance that groups have equal speed is zero. Groups genuinely differ in wait times.
RQ1 No-Reply Shares	$p = 3.34 \times 10^{-20}$	Chance that silence is uniform is zero. Water Districts genuinely ignore citizens.
RQ2 Outcomes by Group	$p = 2.32 \times 10^{-27}$	Chance that disclosure rates are equal is zero. Institutional sector matters.
RQ3 Speed vs. Success	$\boldsymbol{p = 0.989}$	Matches pure random chance. Speed gives ZERO advantage in getting approved.
RQ3 Sensitivity (Water)	$\boldsymbol{p = 0.0164}$	Flips below 0.05! In Water Districts alone, being ignored is directly tied to failure.

The Top 3 Professor Trap Questions & How to Defuse Them:

⚠ PROFESSOR TRAP & SKEPTIC CHALLENGE

Trap 1: "Does slow response cause citizen requests to be denied?"
Your Defense: "No, sir/ma'am! First, this is an observational study, so we cannot assert causality. Second, our data shows the exact opposite: denied requests received faster initial replies (median 1.01d) than approvals (1.51d) due to rapid administrative rejections."

⚠ PROFESSOR TRAP & SKEPTIC CHALLENGE

Trap 2: "Why didn't you measure final resolution time instead of first response time?"
Your Defense: "First, Executive Order No. 2 legally mandates a 15-working-day ceiling for the initial agency response. Second, portal resolution timestamps are frequently overwritten or missing when requesters abandon tickets. The first reply timestamp is immutable in the conversation thread."

⚠ PROFESSOR TRAP & SKEPTIC CHALLENGE

Trap 3: "If your RQ2 p-value is 10^{-27}, doesn't that mean the difference between groups is huge?"
Your Defense: "No! p-value does not measure effect size. With $N = 14,668$, even a 1% difference produces a tiny p-value. That is why we reported **Cramér's $V = 0.096$**, which proves the practical real-world association is weak."

10. Master Statistical & Numerical Reference Cheat-Sheet

Component	Exact Metric	Method / Test	Spoken Meaning / Defense
Full Frame	14,668 requests	Listing census (p. 1024–2002)	Complete 2025 public population on portal
Stratified Sample	1,641 requests	24 strata (Group × Quarter)	Detail pages with minute-level conversation logs

Lost Pages	22 vanished	Excluded; weights adjusted	77% were denied; replacements would cause bias
Response Time Skew	Skew = +4.82	Mean 17.4d vs Median 1.58d	Extreme right skew; required non-parametric tests
Normality Test	Rejected in 6 of 6	Shapiro–Wilk ($p < 10^{-18}$)	Mathematically invalidates One-Way ANOVA
Log Normality Test	Rejected in 6 of 6	Shapiro–Wilk on $\log(x)$ ($p < 0.002$)	Proves transformations cannot rescue ANOVA
RQ1 Test (Speed)	$H = 218.71, p = 2.80 \times 10^{-45}$	Kruskal–Wallis on ranks	Agency groups differ significantly in speed
RQ1 Effect Size	$\varepsilon^2 = 0.152$	$\frac{H}{\frac{N^2-1}{N+1}}$	Moderate practical effect size
RQ1 Post-Hoc	11 of 15 pairs	Mann–Whitney with Bonferroni	LEA (0.56d) fastest; LGU (4.68d) slowest
Institutional Silence	$\chi^2 = 100.94, p = 3.34 \times 10^{-20}$	Chi-Square on no-reply shares	Water Districts ignore 26.1%; LEA ignores 0.7%
RQ2 Test (Outcomes)	$\chi^2 = 134.77, p = 2.32 \times 10^{-27}$	Chi-Square on frame ($N = 14,668$)	Groups differ in disclosure (LGU 75%, Water 33%)
RQ2 Effect Size	Cramér's $V = 0.096$	$\sqrt{\frac{\chi^2}{N \times (k-1)}}$	Weak association; group alone doesn't decide fate
Closed Denominator	$\chi^2 = 123.88, p = 4.73 \times 10^{-25}$	Sensitivity on closed ($n = 12,969$)	Proves RQ2 outcome is robust to open requests
RQ3 Test (Speed vs Yes)	Dominance = 0.500, $p = 0.989$	Weighted Mann–Whitney (9,999)	**Speed has ZERO link to success ("Fast != Yes")**
RQ3 Medians	Success 1.51d vs Fail 1.01d	Weighted median on closed replied	Denials actually received faster replies
Within-Group Speed	All 6 groups $p > 0.05$	Mann–Whitney within each group	No Simpson's paradox; true across every sector
Silent Sensitivity	Water Districts $p = 0.0164$	22 silent closed treated as slowest	In Water Districts, silence guarantees failure
Portal Blackout Gap	March 20 – June 18, 2025	Timeline audit (90 days)	0 public filings; Q2 holds only 324 requests

Prepared for CS 132 Capstone Presentation Defense · Generated November 2026
Repository: `hugvan/cs132-project` · Google Sheet Dataset: `efoi_2025`